

- Q.34 Explain origin of glass.
Q.35 Explain casting of glass.

SECTION-D

- Note:** Long answer type questions. Attempt any two questions out of three questions. (2x10=20)
- Q.36 Explain manufacturing process of glass with flow diagram.
- Q.37 Describe properties and applications of different types of glass.
- Q.38 Explain raw materials used in glass making and their functions.

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4th Sem / Ceramics Engineering Sub : Glass Technology -I

Time : 3Hrs.

M.M. : 100

SECTION-A

- Note:** Multiple choice questions. All questions are compulsory (10x1=10)
- Q.1 Which oxide gives blue colour to glass?
a) Iron oxide b) Cobalt oxide
c) Chromium oxide d) Zinc oxide
- Q.2 Soda in glass acts as
a) Network former b) Flux
c) Stabilizer d) Colouring agent
- Q.3 Which furnace is used for large scale glass production?
a) Pot furnace b) Tank furnace
c) Muffle furnace d) Cupola
- Q.4 Main stabilizer in soda-lime glass is
a) Na_2O b) CaO
c) SiO_2 d) Al_2O_3
- Q.5 The chemical formula of borax is
a) $\text{Na}_2\text{B}_4\text{O}_7$ b) Na_2CO_3
c) Na_2SO_4 d) NaCl

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- Q.6 Glass transition temperature is related to
 a) Density b) Viscosity
 c) Colour d) Hardness
- Q.7 Which impurity in sand causes green colour in glass?
 a) Copper b) Iron
 c) Zinc d) Lead
- Q.8 Which test is used to measure hardness of glass?
 a) Brinell test b) Vickers test
 c) Mohr test d) Charpy test
- Q.9 Which oxide increases refractive index of glass?
 a) PbO b) CaO
 c) Na₂O d) MgO
- Q.10 Glass is mainly composed of
 a) Crystals
 b) Polymer chains
 c) Random network structure
 d) Metal atoms

SECTION-B

Note: Objective type questions. All questions are compulsory. (10x1=10)

- Q.11 Broken glass used in batch is called _____.
 Q.12 Glass shows long range order. (True/False)
 Q.13 Formula of soda ash is _____.

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- Q.14 Alumina increases durability of glass. (True/False)
 Q.15 Refining removes _____ from molten glass.
 Q.16 Silica is acidic oxide. (True/False)
 Q.17 Pot furnace is batch type furnace. (True/False)
 Q.18 Colour of glass depends on metallic _____.
 Q.19 Glass is transparent material. (True/False)
 Q.20 Density of glass is measured in _____.

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

- Q.21 Explain role of silica in glass making.
 Q.22 Explain fluxes used in glass.
 Q.23 Explain stabilizers in glass.
 Q.24 Explain chemical durability of glass.
 Q.25 Explain moisture estimation in glass.
 Q.26 Explain tank temperature control.
 Q.27 Explain optical properties of glass.
 Q.28 Explain conveying of batch materials.
 Q.29 Explain viscosity temperature relationship.
 Q.30 Explain annealing of glass.
 Q.31 Write uses of floaters in glass making
 Q.32 Explain impact strength of glass.
 Q.33 Explain surface tension of glass.

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